

CE 605: Engineering Seismology and Seismic Hazard Assessment

Course Code	CE 605
Course Name	Engineering Seismology and Seismic Hazard Assessment
L-T-P-C	3-0-0-3 Credits Prerequisites: Structural Dynamics with Application to Earthquake Engineering (CE 511) or equivalent course
Intended for	B.Tech. 4th year / M.S./ M.Tech./ Ph.D.
Distribution	Discipline Elective (UG, 4th year); Discipline Elective (PG)
Approval	16th BoA (Feb 2017); 15th Senate (Feb 2018)

Course Contents

- **Introduction:** Internal Structure of the Earth; Plate tectonics and boundaries; Faults; Focal mechanisms; Location of Earthquakes; Size of Earthquakes; Major Earthquakes in the world; Important Indian Earthquakes. [5 Lectures]
- **Wave Propagation in Elastic Medium:** Waves in Infinite Medium; longitudinal and Shear waves; Waves in Semi - infinite medium; Reflection and refraction of waves; Rayleigh waves and Love waves; Response of an elastic Half - Space due to surface and buried forces. Seismic moment tensor. Time dependence. [11 Lectures]
- **Strong Ground Motion and Parameters:** Strong - Motion measurement; International and Local strong motion networks; Synthetic strong ground motions; Stochastic seismological models; Empirical Green's Function method; One - Dimensional Ground response analysis; Ground motion parameters. [12 Lectures]
- **Seismic Hazard Analysis:** Definitions - seismic hazard, disaster and risk; Deterministic seismic hazard analysis; Probability; Earthquake occurrence models; Estimation of maximum magnitude, maximum credible earthquake, design basis earthquake; Ground motion prediction equations; Return periods and strong motion exceedance rates; seismic Hazard curves; Deaggregation. Seismic microzonation, Case studies/Project on estimating the seismic hazard of important Himalayan Cities. [14 Lectures]

Text Books

1. S. L. Kramer, Geotechnical Earthquake Engineering, 2nd Edition, Person Education, 2004.
2. K. F. Graff, Wave Motion in Elastic Solids, Dover Publications, 1991.

References

1. D. M. Boore, Simulation of Ground Motion Using the Stochastic Method, Pure and Applied Geophysics, 160, 635 - 676, 2003.
2. Keiiti AKI and Paul G Richards, Quantitative Seismology, 2nd Edition, University Science Books, 2002.
3. A. Udias, Principles of Seismology, Cambridge University Press, 1999.
4. NEMA, Development of probabilistic seismic hazard map of India, Technical Report, Working Committee of Experts (WCE), National Disaster Management Authority (NOMA), 2011
5. NPTEL course on **Geotechnical Earthquake Engineering** (online resources).